

Assignment 1

March 30, 2026

Problem 1

In this problem, you are provided an opportunity to perform hands-on calculation of the SVM loss and softmax loss we learned in class.

We define a linear classifier:

$$f(\mathbf{x}, \mathbf{W}) = \mathbf{W}\mathbf{x} + \mathbf{b} \quad (1)$$

and provide a data sample:

$$\mathbf{x}_i = \begin{bmatrix} 12 \\ 20 \\ -6 \\ 35 \end{bmatrix}, y_i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \quad (2)$$

Assume that the weights of our model are given by

$$\mathbf{W} = \begin{bmatrix} 0.1 & -0.5 & 0.02 & 0.05 \\ 0.7 & 0.2 & 0.05 & 0.3 \\ 0.0 & -0.04 & -0.2 & 0.03 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 0.0 \\ -0.2 \\ 0.4 \end{bmatrix}. \quad (3)$$

Please calculate 1) SVM loss (hinge loss) and 2) softmax loss (cross-entropy loss) of this sample using the natural log.

Problem 2

The goal of this problem is to minimize a function given a certain input using gradient descent by breaking down the overall function into smaller components via a computational graph. The function is defined as:

$$f(x_1, x_2, w_1, w_2) = \frac{1}{1 + e^{-(w_1 x_1 + w_2 x_2)}} + 0.1(w_1^2 + w_2^2). \quad (4)$$

(a) Please calculate $\frac{\partial f}{\partial w_1}$, $\frac{\partial f}{\partial w_2}$, $\frac{\partial f}{\partial x_1}$, $\frac{\partial f}{\partial x_2}$.

(b) Start with the following initialization: $w_1 = 0.6$, $w_2 = -0.5$, $x_1 = -0.2$, $x_2 = 0.4$. Draw the computational graph. Please use backpropagation as we did in class. You can draw the graph on paper and insert a photo into your report. The goal is for you to practice working with computational graphs. As a consequence, you must include the intermediate values during the forward and backward pass.

Problem 3

Implement a two-layer Multilayer Perceptron (MLP) using **NumPy** with the following specifications:

1. Architecture

- Input layer: d_{in} dimensions
- Hidden layer: h units with ReLU activation
- Output layer: d_{out} units (logits)
- Loss function: **Softmax Cross-Entropy Loss**

2. Requirements

- Initialize weights W_1 (shape (d_{in}, h)) and W_2 (shape (h, d_{out})) with Gaussian random values
- For random input batch X (shape (N, d_{in})) and one-hot labels y (shape (N, d_{out})):
 - Compute forward pass (ReLU activation + softmax loss)
 - Compute gradients of loss w.r.t W_1 and W_2 via backpropagation
- **Prohibited:** Automatic differentiation libraries

3. Deliverables

- Complete the code template
- Print numerical gradients dW_1 and dW_2 (rounded to 4 decimal places)